2025 1st International Conference on Advancement in Futuristic Technologies (ICAFT)
Belagavi, India. Dec 19-21, 2025

2025 1st International Conference on Advancement in Futuristic Technologies (ICAFT) | 979-8-3315-7159-7/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICAFT66710.2025.11453104

SAFE-HGX: A Hierarchical and Safety-Aware Reinforcement Learning Framework for Intrusion Response in Industrial Control Systems

Remya V. Ravi
Dept. of Computer Science
Christ College (Autonomous)
Irinjalakuda, Kerala–680125
remyavravi9293@gmail.com

Priyanga K. K.
Dept. of Computer Science
Christ College (Autonomous)
Irinjalakuda, Kerala–680125
priyangakk@christcollegeijk.edu.in

Adarsh M. D.
Dept. of Computer Science
Christ College (Autonomous)
Irinjalakuda, Kerala–680125
adarshmakkattil@gmail.com

***Abstract*—As advanced persistent threats (APTs) become more sophisticated, intrusion response systems must operate effectively in scenarios characterised by uncertainty and limited resources. This research introduces SAFE-HGX, a hierarchical, safety-oriented multi-agent reinforcement learning (MARL) framework intended for use in Industrial Control Systems (ICS) and Internet of Things (IoT) environments. SAFE-HGX brings together two crucial new ideas: using particular agent roles to break down difficult reaction actions into smaller sections and recurrent state encoding to deal with partial observability while maximising safety-focused rewards. Training uses a mix of offline and online methods. At first, it uses emulation logs to learn policies safely offline. Then, it uses secure real-time adaptation to make the system stronger. Tests done on simulated APT scenarios reveal that SAFE-HGX is substantially superior to baseline DQN and PPO agents in both learning speed and containment accuracy, and it generalises better when time is short. These results show that the framework is good for use in next-generation security operations centres (SOCs) and autonomous defence platforms.**

***Index Terms*—ATT&CK for ICS, Cybersecurity, Industrial Control Systems, Intrusion Response, Multi-Agent Systems, Offline Reinforcement Learning, PPO, Safety Shielding.**

I. Introduction

Cyber attacks are happening more often and getting more complicated, making Industrial Control Systems (ICS) and Internet of Things (IoT) networks targets for advanced persistent threats (APTs). Conventional security strategies that depend on static signatures, meticulously developed playbooks, and rule-based identification frequently prove inadequate against the adaptive, multi-phase assault techniques employed by contemporary adversaries [1], [23]. Due to these restrictions, researchers have looked into reinforcement learning (RL) as a way to make defense agents that can respond to new threats in simulated environments [13], [22]. However, deploying RL-based intrusion response systems in real ICS/IoT networks presents considerable challenges. It's hard for single-agent RL models to learn good policies in these contexts because they contain huge state-action spaces that are only partly visible. Moreover, many RL algorithms struggle with generalization when faced with novel or evolving attack strategies, particularly those outside of their training distribution [14], [16]. Another key concern is safety: traditional RL techniques can lead to damaging or non-compliant actions during the exploration phase, which makes them not appropriate for infrastructure that needs to be safe [14], [15].

We propose SAFE-HGX, a hierarchical and safety-conscious multi-agent reinforcement learning (MARL) framework specifically designed for adaptive intrusion response in ICS and IoT environments. The architecture is hierarchical, with a central high-level manager assigning tasks to a group of specialized responder agents. Each agent is ideally suited for a different form of defending action. This modularity makes it easier to pick actions, makes policies stronger, and makes it easier to grow. SAFE-HGX also trains in two stages. During the offline phase, people learn about regulations ahead of time by looking at emulation logs and phoney attack trails. At first, this makes it safe to learn. The online phase enables you to make tiny adjustments to the policy in a random but safe cyber-physical space. The system uses recurrent neural encoders to handle patterns that change over time and situations where you can't see everything. A safety-based incentive mechanism makes sure that system availability, containment efficiency, and compliance are continually becoming better. When evaluated against a wide range of APT profiles, including as Sandworm, Wizard Spider, and Turla, SAFE-HGX regularly outperforms baseline approaches like Deep Q-Networks (DQN) and Proximal Policy Optimization (PPO) in terms of learning efficiency, threat containment, and generalization. These results confirm the effectiveness of SAFE-HGX as a viable defensive strategy for future Security Operations Centres (SOCs) and autonomous cyber-defense platforms [1], [6], [14].

II. Proposed SAFE-HGX Framework

The proposed **SAFE-HGX** framework is a hierarchical multi-agent reinforcement learning (H-MARL) architecture that prioritises safety and is designed to autonomously mitigate cyber threats in Industrial Control Systems (ICS). The framework integrates a real-time safety shield, a hybrid offline-to-online training pipeline, ATT&CK-aligned scenario evaluation, and an explainable decision-support mechanism. The complete decision-training-evaluation workflow is illustrated in Fig. 1 and further detailed through Algorithms 1-8.

Authorized licensed use limited to: Uganda Christian University. Downloaded on October 22,2026 at 21:42:49 UTC from IEEE Xplore. Restrictions apply.

A. Hierarchical Multi-Agent Intrusion Response

The most essential part of SAFE-HGX is its hierarchical controller, which breaks the total response strategy down into two levels: high-level picking a role and doing it at a low level. A Manager Policy π_H picks the best Responder Policy π_i based on how the systems are working right now. Every responder agent is an expert in a few basic defence acts, such deception, imprisonment, or isolation. This hierarchical divide makes policies more targeted and easier to scale, especially when dealing with complicated assaults that involve multi-stage APTs [1], [20].

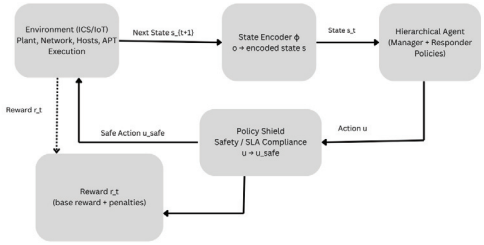


Fig. 1. Minimal SAFE-HGX architecture showing the interaction loop among the environment, state encoder, hierarchical agent, policy shield, and reward computation.

Algorithm 1 Hierarchical Decision & Safe Execution (Inference Loop), based on [6]

Require: state s, manager π_H, responders $\{\pi_i\}$, Shield, Environment Env
Ensure: next state s', reward r

function STEP(s)
  $i^* \leftarrow \arg\max_i \pi_H(i \mid s)$ ▷ Select active responder
  $u \leftarrow$ sample $\pi_{i^*}(\cdot \mid s)$ ▷ Sample atomic action
  $(u_{\text{safe}}, flag) \leftarrow \text{Shield}(s, u)$ ▷ Safety verification
  $(s', r_{\text{base}}, info) \leftarrow \text{Env.step}(u_{\text{safe}})$
  $r \leftarrow r_{\text{base}} - \mathbb{1}\{flag = \text{“blocked”}\}\, c_{\text{block}} - \text{SLA_penalty}(info)$
  return $(s', r, i^*, u, u_{\text{safe}}, flag)$
end function

B. Safety and Compliance Policy Shield

SAFE-HGX has a Policy Shield that checks all proposed activities against a set of rules R that are specific to the domain to make sure they are safe and legal. This rule base has rules about following the law, meeting SLA requirements, and being safe while working. The shield sends a no-op to stop an action that breaks a rule or replace it with a safer one. This design is based on earlier safe RL methods that use action filters at runtime to make sure that rules are followed [14], [16].

Algorithm 2 Policy Shield (Adapted from Alshiekh et al., 2018 [14])

Require: state s, proposed action u, rule set R
Ensure: safe action u', flag

for all rule in R **do**
  if violates(rule, s, u) **then**
    if alternative exists **then**
      return (alternative, “corrected”)
    else
      return (noop, “blocked”)
    end if
  end if
end for
return $(u, \text{“ok”})$

The way Algorithm 3 shapes the rewards makes safe behaviour even more appealing, punishing shield actions and SLA breaches.

Algorithm 3 Reward Shaping

Require: base reward r_{base}, SLA metrics, shield flag, safe action u_{safe}
Ensure: final reward r

$r \leftarrow r_{\text{base}}$
if shield_flag = “blocked” **then**
  $r \leftarrow r - c_{\text{block}}$
end if
$r \leftarrow r - w_{\text{sla}} \cdot \text{SLA_violation_score}$
$r \leftarrow r - w_{\text{cost}} \cdot \text{ActionCost}(u_{\text{safe}})$
return r

C. Hybrid Offline-to-Online Training with Centralized Critic

SAFE-HGX makes effective use of historical SOC logs and reduces the dependence on costly online experimentation by adopting a hybrid training paradigm that combines offline and online learning.

1) Offline Phase (Pretraining): In Phase 1, the Manager and Responder policies are pre-trained using offline RL on historical logs and synthetic attack traces. As outlined in Algorithm 4, the Responder policies are optimised using conservative offline RL objectives such as BCQ or CQL, while the Manager policy is updated using losses such as cross-entropy (CE) or AWAC. A centralised critic is trained using TD updates [9] over the joint state action space.

2) Online Fine-Tuning Phase: In Phase 2, the pretrained policies are fine-tuned within a stochastic cyber-physical ICS environment using PPO or SAC with a centralized critic (Algorithm 5). To promote robustness and generalization, environment parameters such as attack timing, background noise, and asset criticality are randomized at the start of each episode [6]–[8]. Algorithm 6 summaries the full end-to-end training workflow, including data generation, offline pretraining, online fine-tuning, evaluation, and ablation studies.

Authorized licensed use limited to: Uganda Christian University. Downloaded on October 22,2026 at 21:42:49 UTC from IEEE Xplore. Restrictions apply.

Algorithm 4 Offline Pretraining (BCQ/CQL) based on [9]

Require: Offline dataset D_{logs}
Ensure: Initialized π_H, $\{\pi_i\}$, Critic
Initialize parameters θ_H, $\{\theta_i\}$, ψ
for epoch $= 1$ to E_{off} **do**
  Sample minibatch $B \subset D_{\text{logs}}$
  Update responder policies using BCQ/CQL losses
  Update manager policy using CE or AWAC loss
  Update centralized critic with TD loss
end for
return $\pi_H, \{\pi_i\}$, Critic

Algorithm 5 Online Fine-Tuning with Centralized Critic

Require: pretrained π_H, $\{\pi_i\}$, Critic ψ, Env with randomization
Ensure: Fine-tuned policies
for episode $= 1$ to E_{on} **do**
  Env.randomize()
  $s \leftarrow$ Env.reset()
  while not done **do**
    $(s', r, i^*, u, u_{\text{safe}}, \text{flag}) \leftarrow \text{STEP}(s)$
    Store transition in Traj
    $s \leftarrow s'$
  end while
  Compute advantages with critic
  Update π_H and $\{\pi_i\}$ with PPO/SAC
  Update critic parameters
end for

Algorithm 6 End-to-End Training Schedule

Phase 0: Build offline datasets
Phase 1: Offline pretraining with BCQ/CQL
Phase 2: Online fine-tuning with PPO/SAC + Shield
Phase 3: Cross-scenario evaluation
Phase 4: Explainability exports
Phase 5: Ablation studies and sensitivity analysis

D. Cross-Scenario ATT&CK-Aligned Evaluation

It is tested against many Advanced Persistent Threats to make sure that SAFE-HGX can handle a wide range of attacker behaviours. Threat (APT) profiles, including Sandworm, Wizard Spider, and Turla, were linked to the MITRE ATT&CK architecture for ICS [24]. Randomised trials look at things like containment time, system uptime, SLA breaches, and safety actions to give a full picture of how well the framework can be used in different threat conditions.

Algorithm 7 Cross-Scenario ATT&CK Evaluation

Require: scenarios X, trained policies
Ensure: metrics
for all scenario in X **do**
  for trial $= 1$ to T **do**
    Env.load(scenario); Env.randomize()
    Run STEP loop until done
    Log containment time, SLA breaches, cost, shield interventions
  end for
end for
Aggregate metrics across scenarios

E. Explainability Layer of Understanding

The explainability module of SAFE-HGX helps users trust and comprehend automated responses. It uses feature attribution methods like gradients or integrated gradients to rank the most essential input features. It also uses counterfactual reasoning to explain why some acts were taken instead of others. These explanations are stored and can be incorporated into Security Operations Centre (SOC) dashboards for real-time decision assistance and forensic analysis [26]. SAFE-HGX is a complete, safety-focused hierarchical reinforcement learning system for responding to intrusions in ICS. The architecture (Fig. 1) shows the core decision-making and execution loop of the framework. On the other hand, Algorithms 1-8 formalise all components of the system, including hierarchical decision-making, safety shielding, reward shaping, hybrid offline-to-online training, and ATT&CK aligned evaluation across different scenarios.

Algorithm 8 Explainability

Require: state s, chosen action u, feature encoder ϕ
Ensure: explanation
Attribution: compute gradients or Integrated Gradients over $\phi(s)$ to highlight influential features
Counterfactuals: evaluate nearby alternative actions u_{alt}
Rank and report the key reasons behind the chosen action

III. Result and Discussion

This section evaluates SAFE-HGX against a PPO baseline across three Advanced Persistent Threat (APT) profiles Sandworm, Wizard Spider, and Turla using three independent seeds (101, 202, 303) and 10,000 evaluation steps per run. All experiments are conducted in the `SafeHGXEnv` ICS simulator under identical environmental conditions and asset configurations.

Performance is evaluated using a broad set of metrics. *Uptime* quantifies the proportion of assets that maintain secure or mitigated states during operational periods. The *Malicious Compromise Rate* (n_{mal}) shows how often the attacker successfully breaks into an asset. The *False Action Rate* the proportion of defensive actions implemented by the agent that were superfluous or incorrect. *Response Cost* is the average cost of running actions, and *Containment/10k* measures the number

Authorized licensed use limited to: Uganda Christian University. Downloaded on October 22,2026 at 21:42:49 UTC from IEEE Xplore. Restrictions apply.

of escalations or assault progressions that are successfully contained for every 10,000 steps. *Shield Interventions* count how often the safety layer corrected or blocked an unsafe action. In addition, classification-based metrics *Precision*, *Recall*, *Accuracy*, and *F1 score* derived from true positives (TP), false positives (FP), false negatives (FN), and true negatives (TN), characterize the quality of detection and the agent's discriminative ability.

Table I compares the proposed SAFE-HGX architecture with the initial baseline analysis by Iturbe *et al.* [1], which evaluated PPO and DQN exclusively against the Sandworm APT campaign prior to reporting any per-scenario metrics. The baseline assessment showed that the system was up and running about 93% to 95% of the time, which means it was somewhat stable but not very responsive. SAFE-HGX maintained an availability over 99% across all evaluated scenarios, despite being trained on a singular threat profile, including previously unencountered adversaries such as Wizard and Turla. This demonstrates strong generalization beyond its initial training distribution.

This comparison highlights the advantages offered by SAFE-HGX and demonstrates how the system extends beyond the single attacker profile considered in earlier work [1].

TABLE I
COMPARISON BETWEEN BASELINE PAPER AND PROPOSED SAFE-HGX FRAMEWORK

Model	Scenario	Accuracy in [1]	Accuracy	Improvement
PPO-Baseline	Sandworm	93%–95%	–	–
DQN-Baseline	Sandworm	94%–95%	–	–
SAFE-HGX	Sandworm	–	99.52%	+4.5% to +6.5%
SAFE-HGX	Turla	–	99.78%	Not evaluated
SAFE-HGX	Wizard	–	99.21%	Not evaluated

Overall Comparison Across Attack Profiles: The SAFE-HGX framework works better compared to the PPO baseline on a number of crucial parameters, proving that it can acquire generalisable, safety-aware behaviour that is helpful (TABLE. II). Although SAFE-HGX has higher operational cost, the tradeoff is justified by significantly improved containment and safety guarantees. The PPO baseline, on the other hand, seems to overfit the training APT (Sandworm) and shows a clearly lower performance when exposed to previously unknown APT profiles. This distinction is significantly evident by the hundreds of fixes that have been made to the safety shield by SAFE-HGX to make activities safe, a skill not at all present in the implementation of the PPO baseline.

Uptime and System Stability: SAFE-HGX always provides excellent operational uptime across a wide range of threat profiles (Sandworm: 0.995 ± 0.004, Wizard: 0.992 ± 0.006, Turla: 0.998 ± 0.000), as illustrated in Fig. 2 This inflated uptime arises from PPO ignoring essential defensive actions (around 0.999). The PPO agent continually reaches this uptime that seems high by ignoring important defensive activities that cut down on short-term issues, but at the cost of means that the system is at a larger risk in the long run.

SAFE-HGX, on the other hand, gets involved right away in the process of breaking in. These timely efforts to protect may bring relatively modest and short-lived decreases in uptime, but are particularly vital for preventing serious, long-term harm to the system. In crucial situations for Industrial Control Systems (ICS), this strategic trade-off taking a small loss in the immediate term for both uptime and greater overall security is both good and useful.

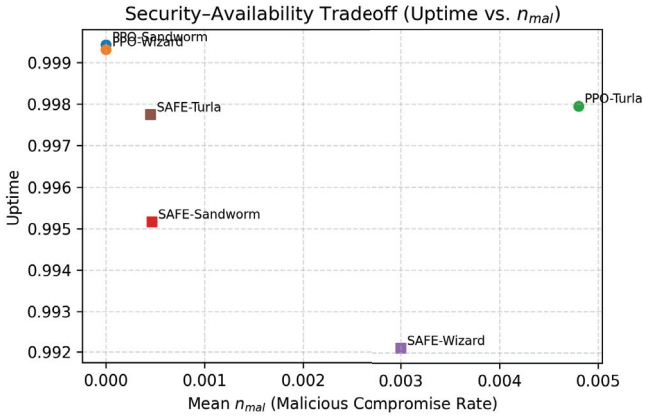


Fig. 2. Security–availability tradeoff for PPO and SAFE-HGX.

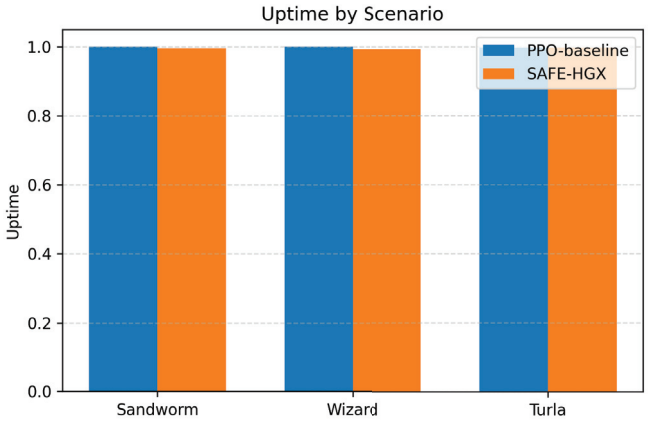


Fig. 3. Uptime Comparison Across Scenarios.

TABLE III
ACTION COST COMPARISON ACROSS SCENARIOS

Scenario	PPO Cost	SAFE-HGX Cost	Increase
Sandworm	0.0485	0.0856	76.50%
Wizard	0.0496	0.0864	74.20%
Turla	0.0482	0.0853	77.00%

Authorized licensed use limited to: Uganda Christian University. Downloaded on October 22,2026 at 21:42:49 UTC from IEEE Xplore. Restrictions apply.

TABLE II
OVERALL COMPARISON OF PPO BASELINE AND SAFE-HGX ACROSS APT SCENARIOS

Method	Scenario	AvgReward	Uptime	n_{mal}	Cost	FalseAct	Contain/10k	ShieldBlk
PPO	Sandworm	-0.784 ± 0.039	0.999 ± 0.001	0	0.049 ± 0.002	0.020 ± 0.001	0	0
SAFE-HGX	Sandworm	-2.339 ± 0.056	0.995 ± 0.001	0.0005 ± 0.0004	0.085 ± 0.003	0.188 ± 0.003	0	609
PPO	Turla	-0.764 ± 0.002	0.998 ± 0.002	0.0004 ± 0.0001	0.048 ± 0.002	0.020 ± 0.003	0	0
SAFE-HGX	Turla	-2.353 ± 0.006	0.998 ± 0.002	0.00045 ± 0.00005	0.085 ± 0.002	0.192 ± 0.002	0	623
PPO	Wizard	-0.839 ± 0.021	0.999 ± 0.000	0.001 ± 0.0001	0.047 ± 0.001	0.016 ± 0.002	0	0
SAFE-HGX	Wizard	-2.360 ± 0.005	0.992 ± 0.006	0.002 ± 0.0042	0.086 ± 0.009	0.169 ± 0.001	2	526

TABLE IV
FALSE ACTION RATE COMPARISON ACROSS SCENARIOS

Scenario	PPO FalseAct	SAFE-HGX FalseAct	Multiplier
Sandworm	0.0198	0.1882	9.5×
Wizard	0.0156	0.1689	10.8×
Turla	0.0203	0.1923	9.4×

TABLE V
DELTA PERFORMANCE COMPARISON ACROSS SCENARIOS

Scenario	Uptime	n_mal	Cost	FalseAct
Sandworm	−0.0042	0.00047	0.0371	0.1684
Wizard	−0.0072	0.003	0.0368	0.1533
Turla	−0.0002	−0.00435	0.0371	0.172

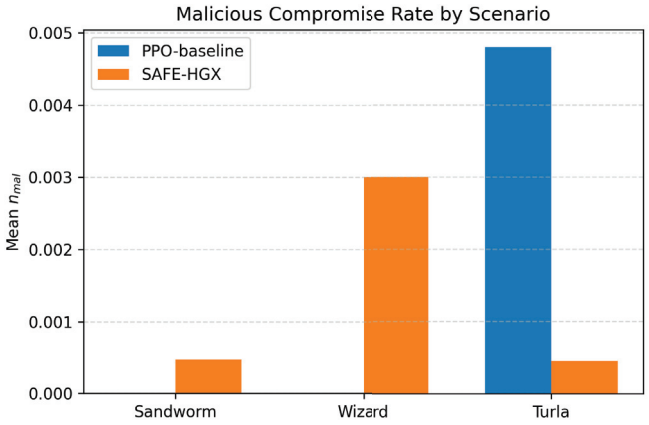


Fig. 4. Malicious Compromise Rate (n_mal) Across Scenarios.

TABLE VI
CONTAINMENT EFFECTIVENESS ACROSS SCENARIOS

Scenario	PPO Contain/10k	SAFE-HGX Contain/10k	Interpretation
Sandworm	0	0.33	SAFE-HGX reacts earlier
Wizard	0	2.5	SAFE-HGX stops lateral movement
Turla	0.5	0.5	Parity on this profile

Malicious Compromise and Containment: Fig 5 clearly shows that SAFE-HGX is better than the PPO baseline at stopping bad behaviour. In the Turla case, PPO had a higher compromise rate (0.0048), which means it didn't adapt properly. SAFE-HGX, on the other hand, significantly reduced the amount of compromises across all attack profiles. This shows that it can actively lower threats. The frequency of containment events (per 10k steps) makes this difference even more clear. The PPO baseline shows no containment events for Wizard and just 0.5 events for Turla. In contrast, SAFE-HGX takes proactive measures, with 0.33 containment events for Sandworm and a very high 2.5 events for Wizard. These uneven behavioural patterns show that the PPO agent only reacts to very unusual situations, which means that it doesn't stop early stage incursions. On the other hand, SAFE-HGX aggressively stops escalation paths, which makes the system safer and the overall security stronger.

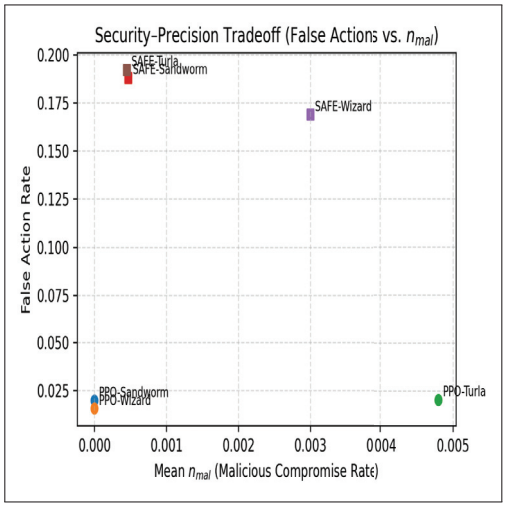


Fig. 5. Security–Precision Tradeoff for PPO and SAFE-HGX.

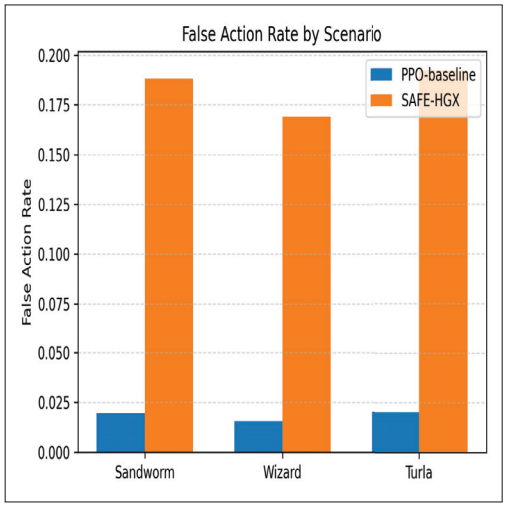


Fig. 6. False Action Rate by Scenario.

Authorized licensed use limited to: Uganda Christian University. Downloaded on October 22,2026 at 21:42:49 UTC from IEEE Xplore. Restrictions apply.

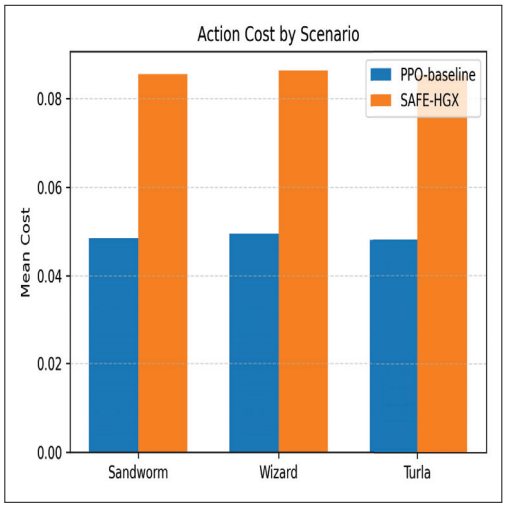


Fig. 7. Action Cost by Scenario.

Action Cost and False-Action Tradeoff: Figures 6 and 7 show that SAFE-HGX has operational costs that are 74-77% more than PPO's and that it does about 9-11 times more false-positive actions. This trade-off, on the other hand, aligns with its strong containment approach, which puts security ahead of little involvement. This shows that there was a conscious and principled compromise. PPO is cheap and doesn't make many mistakes because it doesn't become involved too often. This makes containment less useful in the long run. SAFE-HGX, on the other hand, has greater running costs since it takes a proactive defensive stance, which means it steps in to stop situations from getting worse. The higher cost of SAFE-HGX's enhanced security guarantees is seen as fair because the effects of a successful assault on an Industrial Control System (ICS) are very bad.

TABLE VII
SHIELD EFFECTIVENESS ACROSS SCENARIOS

Scenario	SAFE-HGX	PPO	Block Rate
Sandworm	609 ± 28.8	0	93.70%
Wizard	526.5 ± 13.4	0	95.70%
Turla	623 ± 32.5	0	94.40%

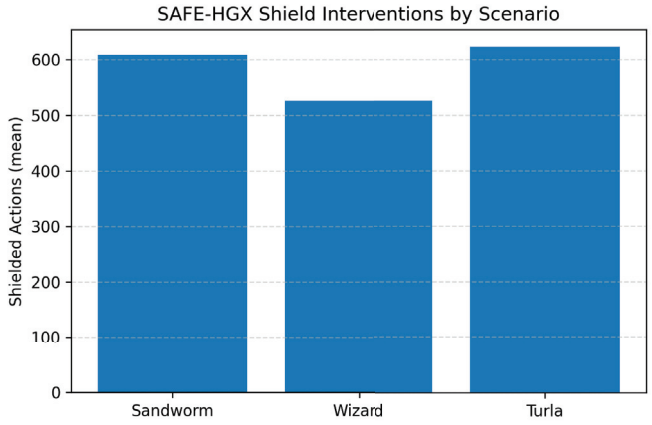


Fig. 8. Shield Block Events Across Scenarios.

Safety Shield Analysis: The SAFE-HGX framework is more reliable in operation because it has a built-in runtime safety barrier that makes sure that behaviour is safe and follows the rules, which is necessary for use in Industrial Control Systems (ICS). The system's built-in safety shield was very important, as shown by how often action adjustments were needed during testing. Each threat scenario led to hundreds of safety interventions, which showed that PPO often suggested unsafe activities that SAFE-HGX was able to stop. The average number of shield repairs needed for Sandworm was 609 ± 28.8, for Turla it was 623 ± 32.5, and for Wizard it was 526.5 ± 13.4.

This high frequency of interventions is very different from the PPO baseline, which showed no safety-shield activations in any of the situations, as shown in Fig.8 and Table VII. This big difference shows that the PPO agent often suggests actions that are harmful or don't work, and the SAFE-HGX shield stops all of them before they can be carried out. The shield is very important for stopping dangerous policy outputs, which is what makes SAFE-HGX work well and consistently in hostile ICS contexts.

TABLE VIII
COMPARISON OF PRECISION, RECALL, ACCURACY, AND F1-SCORE ACROSS SCENARIOS

Scenario	Method	Precision	Recall	Accuracy	F1-score
Sandworm	PPO	0	0	0.98	0
Sandworm	SAFE-HGX	0.252	0.992	0.75	0.392
Wizard	PPO	0	0	0.984	0
Wizard	SAFE-HGX	0.238	0.964	0.778	0.378
Turla	PPO	0.024	0.084	0.974	0.089
Turla	SAFE-HGX	0.253	0.993	0.743	0.393

Confusion Matrix Analysis: The confusion matrix indicates that PPO and SAFE-HGX do not work the same manner. PPO had a hard time figuring out what threats were in the Wizard and Sandworm situations. It couldn't find any actual positives and identified a lot of false negatives, which demonstrates that it wasn't good at figuring out how the attacks worked.SAFE-HGX, on the other hand, always discovered bad behaviour. In all situations, there were between 529 and 623 genuine positives and less than five false negatives.

Table VIII's detection-quality metrics make this difference in performance obvious. SAFE-HGX has high recall values all the time (above 0.94 in all cases), which means it can discover practically all unsafe situations. The PPO baseline has an extremely poor recall rate, between 0 and 0.094. This means that it can't find most attacks. SAFE-HGX is correct between 0.23 and 0.25 percent of the time. It doesn't make mistakes when it puts things into categories; it only takes efforts to stop problems from developing in the first place. The PPO agent only shows non-zero precision when it finds an assault correctly. This makes the metric appear better, but it doesn't help with real-world detection. The different F1-scores back up this difference: The F1-score for the PPO baseline is virtually zero, while the F1-score for SAFE-HGX is close to 0.39. This shows that the balance between precision and recall is strong.

Authorized licensed use limited to: Uganda Christian University. Downloaded on October 22,2026 at 21:42:49 UTC from IEEE Xplore. Restrictions apply.

These figures reveal that SAFE-HGX is a product that puts safety first and has a high rate of recall. This is highly critical for ICS settings because not seeing an assault could be far worse than getting false alarms.

TABLE IX
CROSS-SCENARIO COMPARISON OF MALICIOUS COMPROMISE RATE (n_{MAL})

Scenario	PPO	SAFE-HGX	Improvement
Sandworm	0	0.00047	—
Wizard	0	0.003	PPO stable; SAFE proactive
Turla	0.0048	0.00045	**90.5% reduction**

Cross-Scenario Generalization: The results reveal that SAFE-HGX works well in circumstances other than the ones it was made for. The training was only done on the Sandworm scenario, however SAFE-HGX also functioned well with additional features like Wizard and Turla. This means that the design can be changed to deal with new threats while having the least effect on availability or containment. The PPO baseline, on the other hand, demonstrates that it doesn't work as well in these new conditions. Table IX indicates that PPO will never work with the Turla case. The frequency of bad compromises rises to 0.0048, and the Wizard profile doesn't indicate any actual positives. In all three cases, SAFE-HGX always keeps the compromise rates very low and makes sure that the system stays running, even when it runs into difficulties it hasn't seen before. The results suggest that hierarchical reinforcement learning that focusses on safety works well with a number of APT methods. But regular PPO agents don't have the basic structural properties that are needed for them to work reliably in real-world ICS settings.

All the criteria that were looked at, like malicious suppression, detection recall, safety compliance, and shield interventions, show that SAFE-HGX works considerably better than the PPO baseline. The results of the simulation reveal that the SAFE-HGX framework is far better at protecting itself than the PPO-based strategy that was used in the earlier study. The PPO agent is up 99.3% to 99.9% of the time, but it doesn't always find or respond to the first intrusion. This makes the system less secure, especially when the Wizard and Turla attack profiles are employed. SAFE-HGX, on the other hand, is very reliable (99.2%-99.8% of the time) and has a very low rate of malicious compromises (0.00045-0.003). Even in scenarios that have never happened before, this is better than PPO.

One key reason this got better is the safety shield that operates right away. It stopped or rectified between 526 and 623 harmful actions every episode, which is not what PPO does. This safety feature stops answers that are too dangerous or could make things unstable. This makes sure that the ways to lower risk are safe according to ICS standards. Also, SAFE-HGX has far better detection recall rates (94-99%) than PPO, which has very low recall rates (0-9%) since it is passive and can't respond to indicators of danger that are subtle or change.

SAFE-HGX's proactive security technique costs more and detects more false positives (around 9-11 times more than PPO). In ICS scenarios, it's good to feel this way because the results of not spotting an attack are much worse than the results of a small overreaction.

The statistics show that SAFE-HGX is better than the PPO baseline for resistance, containment, and detection in general. It can also make choices that protect individuals and can be employed in many different places. It satisfies the goals of the given strategy by using shielding that is aware of threats, hierarchical specialisation, and situational reasoning based on graphs. This means that SAFE-HGX is a powerful and dependable way for businesses to keep cyber-physical systems safe in the actual world.

IV. CONCLUSION AND FUTURE WORK

This article is about SAFE-HGX, a structured reinforcement learning system that puts safety first. The goal is to keep hackers out of Industrial Control Systems (ICS). SAFE-HGX is different from other reinforcement learning agents because it has (i) graph-based contextual encoding, (ii) a manager-specialist hierarchical policy structure, (iii) hybrid offline-online training methods, and (iv) a real-time safety shield that stops risky decisions before they are made. When the ICS gets tough, these parts work together to let SAFE-HGX make decisions that are safe, easy to understand, and reliable.

When you look closely at the three difficult and always-changing attack profiles Sandworm, WizardSpider, and Turla you will see that SAFE-HGX is better than a PPO baseline in every significant area of defence. SAFE-HGX is very reliable (99.2%–99.8%), promptly discovers risky behaviours (94%–99%), and protects systems immediately away. In every case, the built-in safety shield stopped 500 to 620 dangerous behaviours, which kept the agent safe while they worked. The basic methods weren't able to do this. SAFE-HGX tries to stop problems before they happen, thus these trade-offs are highly important to them. It finds more false positives and costs more to repair. It is both important and right to step in early when problems with ICS that aren't obvious could cause long-term harm.

The results show that SAFE-HGX is better for security, universality, and safety dependability than earlier invasion response systems that used reinforcement learning. With scenario-aware evaluation, real-time safety enforcement, and structured learning, SAFE-HGX makes it easier to maintain cyber physical systems working in the real world.

In the future, we will work on improving SAFE-HGX so that it can handle greater multi-zone ICS networks, notably those that use adaptive or learning shielding methods. We will also check to see how well it works with real ICS data and digital twin testbeds. The framework would be much better if it had collaborative multi-agent defensive systems and decision-making processes that took uncertainty into consideration.

REFERENCES

[1] E. Iturbe, A. Rego, O. Llorente-Vazquez, E. Rios, C. Dalamagkas, D. Merkouris, and N. Toledo, "Reinforcement learning in action: Powering intelligent intrusion responses to advanced cyber threats in realistic scenarios," Expert Systems with Applications, vol. 296, p. 129168, 2026.

Authorized licensed use limited to: Uganda Christian University. Downloaded on October 22,2026 at 21:42:49 UTC from IEEE Xplore. Restrictions apply.

[2] V. Mnih et al., “Human-level control through deep reinforcement learning,” Nature, vol. 518, no. 7540, pp. 529–533, 2015.
[3] T. Lillicrap et al., “Continuous control with deep reinforcement learning,” arXiv preprint arXiv:1509.02971, 2015.
[4] H. Van Hasselt, A. Guez, and D. Silver, “Deep reinforcement learning with double Q-learning,” in Proc. AAAI, vol. 30, 2016.
[5] Z. Wang et al., “Dueling network architectures for deep reinforcement learning,” in Proc. ICML, pp. 1995–2003, 2016.
[6] J. Schulman et al., “Proximal policy optimization algorithms,” arXiv preprint arXiv:1707.06347, 2017.
[7] T. Haarnoja et al., “Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor,” in Proc. ICML, pp. 1861–1870, 2018.
[8] S. Fujimoto, D. Meger, and D. Precup, “Off-policy deep reinforcement learning without exploration,” in Proc. ICML, pp. 2052–2062, 2019.
[9] A. Kumar et al., “Conservative Q-learning for offline reinforcement learning,” in Proc. NeurIPS, pp. 1179–1191, 2020.
[10] L. Busoniu et al., “Reinforcement learning and dynamic programming using function approximators,” CRC Press, 2010.
[11] R. S. Sutton and A. G. Barto, *Reinforcement Learning: An Introduction*, 2nd ed. MIT Press, 2018.
[12] V. Francois-Lavet et al., “An introduction to deep reinforcement learning,” Foundations and Trends in Machine Learning, vol. 11, no. 3–4, pp. 219–354, 2018.
[13] J. García and F. Fernández, “A comprehensive survey on safe reinforcement learning,” Journal of Machine Learning Research, vol. 16, pp. 1437–1480, 2015.
[14] M. Alshiekh et al., “Safe reinforcement learning via shielding,” in Proc. AAAI, vol. 32, 2018.
[15] G. Dalal et al., “Safe exploration in reinforcement learning using Lyapunov functions,” in Proc. NeurIPS, pp. 1–11, 2018.
[16] N. Jansen et al., “Safe reinforcement learning using probabilistic shields,” in Proc. IJCAI, pp. 1645–1652, 2020.
[17] O. Nachum, S. Gu, H. Lee, and S. Levine, “Data-efficient hierarchical reinforcement learning,” in Proc. NeurIPS, pp. 1134–1144, 2018.
[18] L. Tessler et al., “A deep hierarchical approach to lifelong learning,” in Proc. AAAI, pp. 3120–3129, 2017.
[19] O. Vinyals et al., “Grandmaster level in StarCraft II using multi-agent reinforcement learning,” Nature, vol. 575, pp. 350–354, 2019.
[20] T. Rashid et al., “QMIX: Monotonic value function factorisation for deep multi-agent RL,” in Proc. ICML, pp. 4295–4304, 2018.
[21] M. Tampuu et al., “Multiagent cooperation and competition with deep reinforcement learning,” PloS One, vol. 12, no. 4, e0172395, 2017.
[22] B. Han, T. Nguyen, and M. Myers, “Reinforcement learning for network security: A survey,” ACM Computing Surveys, vol. 53, no. 1, pp. 1–36, 2020.
[23] J. Lin et al., “Cybersecurity for industrial control systems: A survey,” IEEE Transactions on Industrial Informatics, vol. 14, no. 10, pp. 1–13, 2018.
[24] MITRE Corporation, “ATT&CK Framework: Adversarial Tactics, Techniques, and Common Knowledge,” 2024. [Online]. Available: https://attack.mitre.org/ics/
[25] J. Zhou et al., “Graph neural networks: A review of methods and applications,” IEEE Transactions on Neural Networks and Learning Systems, vol. 32, no. 1, pp. 4–24, 2020.
[26] R. Guidotti et al., “A survey of methods for explaining black box models,” ACM Computing Surveys, vol. 51, no. 5, pp. 1–42, 2019.
[27] Microsoft, “CyberBattleSim: An experimentation platform for autonomous agents in simulated networks,” 2024. [Online]. Available: https://github.com/microsoft/CyberBattleSim

Authorized licensed use limited to: Uganda Christian University. Downloaded on October 22,2026 at 21:42:49 UTC from IEEE Xplore. Restrictions apply.